

Displays

D1:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^n,$$

D2:

$$\mathcal{S} = \{f(\boldsymbol{x}) : \boldsymbol{x} \in \mathcal{I}\}.$$

D3:

$$s(d) = \binom{n+d}{n}.$$

D4:

$$\mathbf{M}_d = \int_{\mathbb{R}^n} \mathbf{v}_d(\boldsymbol{x}) \mathbf{v}_d(\boldsymbol{x})^\top d\mu(\boldsymbol{x}).$$

D5:

$$\Lambda_{\mu,d}(\boldsymbol{x}) = \left(\mathbf{v}_d(\boldsymbol{x})^T \mathbf{M}_d^{-1} \mathbf{v}_d(\boldsymbol{x}) \right)^{-1}. \quad (1)$$

D6:

$$\Lambda_{\mu,d}(\boldsymbol{x}) = \inf_{P \in \mathbb{R}[\mathbf{X}]_d^n} \left\{ \int_{\mathbb{R}^n} P(\mathbf{z})^2 d\mu(\mathbf{z}), \quad \text{s.t.} \quad P(\mathbf{x}) = 1 \right\}$$

D7:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \delta_{\mathbf{x}^i},$$

D8:

$$\widehat{\mathbf{M}}_d = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_d(\mathbf{x}^i) \mathbf{v}_d(\mathbf{x}^i)^T \quad (2)$$

D9:

$$\Lambda_{\hat{\mu},d}^{-1}(\boldsymbol{x}) = \mathbf{v}_d(\boldsymbol{x})^T \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\boldsymbol{x}) \quad (3)$$

D10:

$$\hat{\mathcal{S}} = \{\boldsymbol{x} \in \mathbb{R}^n \mid \Lambda_{\mu,d}^{-1}(\boldsymbol{x}) \leq \alpha\} \quad (4)$$

D11:

$$\alpha = \max_i \Lambda_{\hat{\mu},d}^{-1}(\boldsymbol{x}^i). \quad (5)$$

D12:

$$\hat{\mathcal{S}} = \{\boldsymbol{x} \in \mathbb{R}^n \mid \Lambda_{\hat{\mu},d}^{-1}(\boldsymbol{x}) \leq \max_i \Lambda_{\hat{\mu},d}^{-1}(\boldsymbol{x}^i)\}. \quad (6)$$

D13:

$$f(x,y) = (1 + \text{sign}(x) \cdot x^2, 1 + \text{sign}(y) \cdot y^2)$$

D14:

$$\mathcal{S} = [-3, -1]^2 \cup [-3, -1] \times [1, 3] \cup [1, 3] \times [-3, -1] \cup [1, 3]^2.$$

D15:

$$p_{value}(\boldsymbol{x}) = \frac{1}{N} \left| \left\{ i \mid r(\boldsymbol{x}^i) \geq r(\boldsymbol{x}) \right\} \right|$$

D16:

$$C_{\mathcal{D}}^{\frac{i}{N}} = \left\{ \boldsymbol{x} \in \mathbb{R}^n \mid p_{value}(\boldsymbol{x}) \geq \frac{i}{N} \right\}$$

D17:

$$\mathbb{P} \left(\boldsymbol{x}^{N+1} \in C_{\mathcal{D}}^{\frac{i}{N}} \right) \geq 1 - \frac{i+1}{N+1}. \quad (7)$$

D18:

$$\mathbb{P} \left(\boldsymbol{x}^{N+1} \in C_{\mathcal{D}}^{\frac{i}{N}} \mid \mathcal{D}_{\text{cal}} \right).$$

D19:

$$\mathbb{P} \left[U_{(1)} \leq b_1, \dots, U_{(N)} \leq b_n \right] \geq 1 - \delta.$$

D20:

$$\mathbb{P} \left[\mathbb{P} \left(\boldsymbol{x} \in C_{\mathcal{D}_{\text{cal}}}^{\frac{i}{N}} \mid \mathcal{D}_{\text{cal}} \right) \geq 1 - b_i \right] \geq 1 - \delta \quad (8)$$

D21:

$$\mathbb{P}(U_{(N)} \leq \alpha) \geq 1 - \delta.$$

D22:

$$\mathbb{P} \left[\mathbb{P} \left(\mathbf{x} \in C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \mid \mathcal{D}_{\text{cal}} \right) \leq \alpha \right] \geq 1 - \delta \quad (9)$$

D23:

$$\mathbb{P} \left[F_{\mu}(r(\mathbf{x}^N)) \leq \alpha \right] \geq 1 - \delta.$$

D24:

$$\mathbb{P} \left(\mathbf{x} \in C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \mid \mathcal{D}_{\text{cal}} \right) = \mathbb{P} \left(r(\mathbf{x}) \leq r(\mathbf{x}^N) \mid \mathcal{D}_{\text{cal}} \right) = F_{\mu} \left(r(\mathbf{x}^N) \right)$$

D25:

$$\mathbb{P} \left[\mu \left(C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \right) \geq \exp \left(\frac{\log(\delta)}{N} \right) \right] \geq 1 - \delta, \quad (10)$$

D26:

$$\mathbb{P} \left[\exp \left(\frac{\log(1 - \delta)}{N} \right) \geq \mu \left(C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \right) \right] \geq 1 - \delta, \quad (11)$$

D27:

$$\mathbb{P} \left[\exp \left(\frac{\log(1 - \delta)}{N} \right) \geq \mu \left(C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \right) \geq \exp \left(\frac{\log(\delta)}{N} \right) \right] \geq 1 - 2\delta. \quad (12)$$

D28:

$$\mathbb{P} \left(U_{(1)} \leq b_1, \dots, U_N \leq b_N \right) = \mathbb{P} \left(U_{(1)} \leq b_1 \right) = 1 - \mathbb{P} \left(U_{(1)} \geq b_1 \right).$$

D29:

$$1 - \mathbb{P} \left(U_{(1)} \geq b_1 \right) = 1 - \prod_{i=1}^N \mathbb{P} \left(U_i \geq b_1 \right) = 1 - (1 - b_1)^N = 1 - \delta.$$

D30:

$$\mathbb{P} \left[\mathbb{P} \left(\mathbf{x} \in C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \mid \mathcal{D}_{\text{cal}} \right) \geq \exp \left(\frac{\log(\delta)}{N} \right) \right] \geq 1 - \delta.$$

D31:

$$C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} = \hat{\mathcal{S}} = \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{v}_d(\mathbf{x})^\top \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\mathbf{x}) \leq \alpha \right\}$$

D32:

$$r(\mathbf{x}) = \Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}) = \mathbf{v}_d(\mathbf{x})^T \widehat{\mathbf{M}}_x^{-1} \mathbf{v}_d(\mathbf{x}).$$

D33:

$$p_{\text{value}}(\mathbf{x}) = \frac{1}{N} \left| \left\{ i \mid \Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}^i) \geq \Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}) \right\} \right|$$

D34:

$$\Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}) = \frac{\Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x})}{1 + \Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x})}, \quad \Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}^i) = \Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x}^i) - \frac{(\mathbf{v}_d(\mathbf{x})^\top \mathbf{y}^i)^2}{1 + \Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x})}, \quad (13)$$

D35:

$$\mathbb{P} \left(\mu \left(C_{\mathcal{D}}^{\frac{p+1}{N}} \right) \geq 1 - \epsilon \right) \geq \sum_{i=p+1}^{N-p} \binom{N-p}{i} \epsilon^i (1 - \epsilon)^{N-p-i} \quad (14)$$

D36:

$$\begin{aligned} \mathbb{P} \left[U_{(1)} \leq b_1, \dots, U_{(N-m)} \leq b_{N-m} \right] &\geq \mathbb{P} \left[U_{(1)} \leq b_1, \dots, U_{(N-p)} \leq b_{N-p} \right] \\ &\geq \sum_{i=p+1}^{N-p} \binom{N-p}{i} \epsilon^i (1 - \epsilon)^{N-p-i}. \end{aligned}$$

D37:

$$\mu \left(C_{\mathcal{D}_{\text{inliers}}}^{\frac{p+1}{N-m}} \right) = \mathbb{P} \left(\mathbf{x} \in C_{\mathcal{D}_{\text{inliers}}}^{\frac{p+1}{N-m}} \mid \mathcal{D}_{\text{inliers}} \right).$$

D38:

$$\mathbb{P} \left(\mu \left(C_{\mathcal{D}_{\text{inliers}}}^{\frac{p+1}{N-m}} \right) \geq 1 - \epsilon \right) \geq \sum_{i=p+1}^{N-p} \binom{N-p}{i} \epsilon^i (1 - \epsilon)^{N-p-i}$$

D39:

$$\ddot{x} = -\delta \dot{x} + \alpha x - \beta x^3 + \gamma \cos(\omega t)$$

Inline

- I1: $\mathbf{x} \in \mathbb{R}^n$
- I2: f
- I3: $\mathcal{I}$
- I4: $\hat{\mathcal{S}}$
- I5: $\mathcal{S}$
- I6: S
- I7: μ
- I8: $\mathbf{x} \in \mathbb{R}^n$
- I9: $\mathbf{x} = (x_1, \dots, x_n)$
- I10: $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$
- I11: $\mathbf{x}^\alpha = x_1^{\alpha_1} \times x_2^{\alpha_2} \dots \times x_n^{\alpha_n}$
- I12: $d \in \mathbb{N}$
- I13: $\mathbb{R}[\mathbf{X}]_d^n$
- I14: d
- I15: $\boldsymbol{\alpha} \in \mathbb{N}^n$
- I16: $\mathbf{x}^\alpha$
- I17: $\|\boldsymbol{\alpha}\| = \sum_{i=1}^n \alpha_i$
- I18: $\|\boldsymbol{\alpha}\| \leq d$
- I19: $\mathbf{v}_d(\mathbf{x}) \in \mathbb{R}^{s(d)}$
- I20: $\mathbf{x}$
- I21: $d = 2$
- I22: $n = 2$
- I23: $\mathbf{v}_d(\mathbf{x}) = [1 \ x_1 \ x_2 \ x_1 x_2 \ x_1^2 \ x_2^2]$
- I24: $\mathbb{R}^n$
- I25: $\Lambda_{\mu,d}(\mathbf{x}) : \mathbb{R}^n \mapsto \mathbb{R}$
- I26: M_d
- I27: 1
- I28: $N \in \mathbb{N}$
- I29: N
- I30: $\{\mathbf{x}^1, \dots, \mathbf{x}^N\}$
- I31: $\delta_{\widehat{\mathbf{x}}}$
- I32: $\widehat{\mathbf{M}}_d$
- I33: $\hat{\mu}$
- I34: $N \geq s(d)$
- I35: $2d$
- I36: $N > s(d)$
- I37: $\alpha \in \mathbb{R}$
- I38: $\Lambda_{\mu,d}$
- I39: $\mathbf{x}^i$
- I40: α
- I41: $\mathbf{x}^i \subseteq \hat{\mathcal{S}}$
- I42: $\mathcal{D} = \{\mathbf{x}^1, \dots, \mathbf{x}^N\}$
- I43: $N \geq \frac{5}{\epsilon} \left(\log \frac{4}{\delta} + \binom{n+2d}{n} \log \frac{40}{\epsilon} \right)$
- I44: $\mathbb{P}\left(\mu(\hat{\mathcal{S}}) \geq 1 - \epsilon\right) \geq 1 - \delta.$
- I45: N, δ, ϵ
- I46: $1 - \delta$
- I47: $1 - \epsilon$
- I48: $d \rightarrow \infty$
- I49: $n \rightarrow \infty$
- I50: $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
- I51: $\mathcal{I} = [-1, 1]^2$
- I52: $N = 10000$
- I53: ε
- I54: $1 - \delta = 0.99$

I55: $\hat{S}$
 I56: ϵ
 I57: $d = 3$
 I58: $\epsilon = 0.085$
 I59: $d = 6$
 I60: $\epsilon = 0.23$
 I61: $d = 10$
 I62: $\epsilon = 0.51$
 I63: $d = 15$
 I64: $\epsilon = 0.9$
 I65: $r : \mathbb{R}^n \rightarrow \mathbb{R}$
 I66: $\mathcal{D} = \{\mathbf{x}^1, \dots, \mathbf{x}^N\}$
 I67: p
 I68: $i \in \{0, \dots, N\}$
 I69: $\mathbf{x}^{N+1}$
 I70: $\mathcal{D}$
 I71: r
 I72: $\mathcal{D}_{\text{train}}$
 I73: $\mathcal{D}_{\text{cal}}$
 I74: $b_1, \dots, b_n$
 I75: $U_1, \dots, U_N \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([0, 1])$
 I76: $U_{(1)} \leq U_{(2)} \leq \dots \leq U_{(N)}$
 I77: $\delta \in (0, 1)$
 I78: $0 \leq b_1 \leq b_2 \leq \dots \leq b_N \leq 1$
 I79: $b_0 = 0$
 I80: $r(\mathbf{x})$
 I81: F_μ
 I82: $F_\mu(r(\mathbf{x}))$
 I83: $\text{Unif}([0, 1])$
 I84: $F_\mu(r(\mathbf{x}^1)), F_\mu(r(\mathbf{x}^2)), \dots, F_\mu(r(\mathbf{x}^N))$
 I85: $r(\mathbf{x}^1) \leq r(\mathbf{x}^2) \leq \dots \leq r(\mathbf{x}^N)$
 I86: $U_N = F_\mu(r(\mathbf{x}^N))$
 I87: $\blacksquare$
 I88: $\forall \delta \in (0, 1),$
 I89: $\forall \delta \in (0, 1/2)$
 I90: $b_1 \dots, b_N$
 I91: b_1
 I92: $b_2 \dots, b_N = 1$
 I93: $U_1, \dots, U_N \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([0, 1])$
 I94: $b_1 = 1 - \delta^{\frac{1}{N}}$
 I95: $b_2 = \dots = b_N = 1$
 I96: $U_{(1)}$
 I97: $U_1, \dots, U_N$
 I98: $\mathbb{P}(U_{(1)} \geq b_1)$
 I99: U_i
 I100: $\mu \left(C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \right) = \mathbb{P} \left[\mathbf{x} \in C_{\mathcal{D}_{\text{cal}}}^{\frac{1}{N}} \mid \mathcal{D}_{\text{cal}} \right]$
 I101: $\alpha = \exp \left(\frac{\log(1-\delta)}{N} \right)$
 I102: $\mathbb{P}[U_N \leq \alpha] = \alpha^N = 1 - \delta$
 I103: $U_{(N)} \leq \alpha$
 I104: U_i
 I105: $\mathcal{D} = \{\mathbf{x}^1, \dots, \mathbf{x}^M\}$
 I106: $\mathcal{S} = f(\mathcal{T})$
 I107: $N < M$
 I108: $\epsilon = 1 - \delta^{1/N}$
 I109: $M - N$

I110: $\mathcal{D}_{\text{train}} = \{\mathbf{x}^{N+1}, \dots, \mathbf{x}^M\}$
 I111: $\mathcal{D}_{\text{cal}} = \{\mathbf{x}^1, \dots, \mathbf{x}^N\}$
 I112: $\widehat{\mathbf{M}}_d = \frac{1}{M-N} \sum_{i=N+1}^M \mathbf{v}_d(\mathbf{x}^i) \mathbf{v}_d(\mathbf{x}^i)^\top$
 I113: $\mathbf{x}^i \in \mathcal{D}_{\text{train}}$
 I114: $\widehat{\mathbf{M}}_d^{-1}$
 I115: $\alpha = \max_{i=1, \dots, N} \mathbf{v}_d(\mathbf{x}^i)^\top \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\mathbf{x}^i)$
 I116: $\mathbf{x}^i \in \mathcal{D}_{\text{cal}}$
 I117: $M = 10000$
 I118: M
 I119: $N = 2000$
 I120: $1 - \delta =$
 I121: $\epsilon \leq 0.002$
 I122: n
 I123: δ
 I124: $M = 1000$
 I125: $N = 200$
 I126: $\epsilon \leq 0.02$
 I127: 6
 I128: $\epsilon = 0.02$
 I129: $\varepsilon = 0.002$
 I130: $\varepsilon = 0.02$
 I131: $\mathcal{D}_{\text{cal}} := \mathcal{D}$
 I132: $\mathcal{D} = \{\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^N\}$
 I133: $\mathcal{D}_x = \mathcal{D} \cup \{\mathbf{x}\}$
 I134: $\mathcal{D}_x$
 I135: $\hat{\mu}_x$
 I136: $\widehat{\mathbf{M}}_x$
 I137: $\mathcal{O}(s(d)^3)$
 I138: $\Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x})$
 I139: $\Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x})$
 I140: $\mathbf{y}^i = \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\mathbf{x}^i)$
 I141: $\Lambda_{\hat{\mu}, d}^{-1}(\mathbf{x}^i)$
 I142: $\mathbf{y}^i$
 I143: $\mathcal{O}(Ns(d)^2)$
 I144: $\mathcal{O}(Ns(d))$
 I145: $\Lambda_{\hat{\mu}_x, d}^{-1}(\mathbf{x}^i)$
 I146: $\mathcal{O}(s(d))$
 I147: $p_{\text{value}}(\mathbf{x})$
 I148: $\mathcal{O}(Ns(d) + s(d)^2)$
 I149: $M = N = 1000$
 I150: $2p + 1 < N$
 I151: $\epsilon \in (0, 1)$
 I152: $p = 0$
 I153: $\mathcal{D} = \mathcal{D}_{\text{inlier}} \cup \mathcal{D}_{\text{outlier}}$
 I154: $m \leq p$
 I155: $\mathcal{D}_{\text{outlier}}$
 I156: $U_1, \dots, U_{N-m} \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([0, 1])$
 I157: $U_{(1)} \leq U_{(2)} \leq \dots \leq U_{(N-m)}$
 I158: $b_1 = \dots = b_{p+1} = \epsilon$
 I159: $b_{p+2} = \dots = b_{N-p} = \dots = b_{N-m} = 1$
 I160: $\forall m \leq p$
 I161: $0 < i \leq N - p$
 I162: $N - p$
 I163: $V_i, \dots, V_{N-p} \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([0, 1])$
 I164: i

I165: $\binom{N-p}{i} \epsilon^i (1-\epsilon)^{N-p-i}$
 I166: $p+1$
 I167: $\sum_{i=p+1}^{N-p} \binom{N-p}{i} \epsilon^i (1-\epsilon)^{N-p-i}$.
 I168: $C_{\mathcal{D}_{inliers}}^{\frac{p+1}{N-m}} \subseteq C_{\mathcal{D}}^{\frac{p+1}{N}}$
 I169: $\mu\left(C_{\mathcal{D}}^{\frac{p+1}{N}}\right) \geq \mu\left(C_{\mathcal{D}_{inliers}}^{\frac{p+1}{N-m}}\right)$
 I170: $M = 1500$
 I171: $N = 500$
 I172: $\epsilon = 0.15$
 I173: $\mathcal{I} \subset \mathbb{R}^n$
 I174: $M > N$
 I175: $\mathcal{D} = \{\mathbf{x}^{(i)}\}$
 I176: $i \in \{1, \dots, M\}$
 I177: $\mathbf{x}^{(i)}$
 I178: $\mathbf{x}^{(i)} \in f(\mathcal{I})$
 I179: $\sum_{i=p+1}^{N-p} \binom{N-p}{i} \epsilon^i (1-\epsilon)^{N-p-i}$
 I180: $\widehat{\mathbf{M}}_d = \frac{1}{M-N} \sum_{i=N+1}^M \mathbf{v}_d(\mathbf{x}^{(i)}) \mathbf{v}_d(\mathbf{x}^{(i)})^\top$
 I181: $\mathcal{D}_{\text{cal}} = \{\mathbf{x}^{(i)} \mid i \in \{1, \dots, N\}\}$
 I182: $\text{score}_i = \mathbf{v}_d(\mathbf{x}^{(i)})^\top \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\mathbf{x}^{(i)})$
 I183: $i = 1, \dots, N$
 I184: $\text{score}_1 \geq \text{score}_2 \geq \dots \geq \text{score}_N$
 I185: $C_{\mathcal{D}}^{\frac{p+1}{N}}$
 I186: $\hat{\mathcal{S}} = \left\{ \mathbf{x} \in \mathbb{R}^n : \mathbf{v}_d(\mathbf{x})^\top \widehat{\mathbf{M}}_d^{-1} \mathbf{v}_d(\mathbf{x}) \leq \text{score}_{p+1} \right\}$
 I187: $[-4, 4]^2$
 I188: $\hat{S} \setminus S$
 I189: $[-1, 1]^2$
 I190: $\alpha = 1$
 I191: $\beta = 1, \delta = 0.05, \gamma = 0.4$
 I192: $\omega = 1.3$
 I193: $\mathcal{I} = [-0.95, 1.05] \times [-0.05, 0.05]$